

TMA-502**B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022****COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUES****Time : 1½ Hours****Maximum Marks : 50**

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) (i) What is Chopping ? When does it occur ? 5 Marks
(ii) Two numbers x_1 and x_2 are given which are correct upto their last digit : $x_1 = 12.47$, $x_2 = 10.3$. Estimate the maximum absolute error, relative error and percentage error in computing $x_1 - x_2$.

5 Marks (CO1)

OR

- (b) (i) What is machine epsilon ? How is it related to significant digits ? 5 Marks (CO1)
(ii) Obtain the solution of the following system using Gauss-Seidel iteration method : 5 Marks (CO1)

$$2x_1 + x_2 + x_3 = 5$$

$$3x_1 + 5x_2 + 2x_3 = 15$$

$$2x_1 + x_2 + 4x_3 = 8$$

2. (a) (i) Obtain the function whose first difference is : 5 Marks (CO1)

$$5x^4 + 2x^3 - 3x - 10 = 0$$

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(ii) Prove that with usual notation :

5 Marks (CO2)

$$\left(\frac{1}{E^2} + \frac{1}{E^2} \right) (1 + \Delta)^{\frac{1}{2}} = 2 + \Delta$$

OR

(b) (i) Prove that with usual notation :

5 Marks (CO2)

$$\Delta = \frac{\delta^2}{2} + \delta \sqrt{1 + \frac{\delta^2}{4}}$$

(ii) Find the missing terms in the following table :

5 Marks (CO2)

x	y
2	0.135
2.1	?
2.2	0.111
2.3	0.100
2.4	?
2.5	0.082
2.6	0.074

3. (a) Derive the Newton-Raphson's formula using the Taylor series expansion. Hence find the root of the $e^{-x} - \tan x = 0$ correct to four decimal places.

10 Marks (CO1)

OR

- (b) Explain Regula-Falsi method stating that one advantage over bisection method. Also, find the smallest positive real root of $\cosh x \cos x = -1$, using Regula-Falsi method correct upto four decimal places.

10 Marks (CO1)

4. (a) Using Gauss' forward interpolation formula to find the value of y when $x = 3.75$ for the following table : 10 Marks (CO2)

x	$y = f(x)$
2.5	24.145
3.0	22.043
3.5	20.225
4.0	18.644
4.5	17.262
5.0	16.047

OR

- (b) The following are the numbers of death in four successive ten-year age group. Find the number of death at 50-55 and 55-60 : 10 Marks (CO2)

Age Group	Deaths
25—35	13229
35—45	18139
45—55	24225
55—65	31496
65—75	45196

5. (a) Using Newton's divided difference formula, find the value of $\log_{10} 301$, from the following table : 10 Marks (CO2)

x	$y = \log_{10} x$
300	2.4771
304	2.4829
305	2.4843
307	2.4871